End-Sem 2024 (PH3101)

SOLUTIONS

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① $\cancel{\left[\frac{df}{dt}, g\right]}$ $\frac{d}{dt}[f, g]$

$$= \frac{d}{dt}\left(\frac{\partial f}{\partial q}\frac{\partial g}{\partial p} - \frac{\partial f}{\partial p}\frac{\partial g}{\partial q}\right).$$

$$= \frac{d}{dt}\left(\frac{\partial f}{\partial q}\right)\frac{\partial g}{\partial p} + \frac{d}{dt}\left(\frac{\partial g}{\partial p}\right)\cdot\frac{\partial f}{\partial q} - \frac{d}{dt}\left(\frac{\partial f}{\partial p}\right)\frac{\partial g}{\partial q} - \frac{d}{dt}\left(\frac{\partial g}{\partial q}\right)\frac{\partial f}{\partial p}$$

$$\frac{d}{dt}\left(\frac{\partial f}{\partial q}\right) = \frac{\partial^2 f}{\partial^2 q}\cdot \cancel{\partial q}\,\dot{q} + \frac{\partial^2 f}{\partial p \partial q}\dot{p} + \frac{\partial^2 f}{\partial t \partial q}$$

$$= \frac{\partial}{\partial q}\left(\frac{\partial f}{\partial q}\cdot\dot{q} + \frac{\partial f}{\partial p}\dot{p} + \frac{\partial f}{\partial t}\right) \quad ①$$

$$= \frac{\partial}{\partial q}\dot{f} \quad \text{—①}$$

∴ Using ① for both f & g:

$$\frac{d}{dt}[f, g] = \frac{\partial \dot{f}}{\partial q}\frac{\partial g}{\partial p} + \frac{\partial \dot{g}}{\partial p}\frac{\partial f}{\partial q} - \frac{\partial \dot{f}}{\partial p}\frac{\partial g}{\partial q} - \frac{\partial \dot{g}}{\partial q}\frac{\partial f}{\partial p}$$

$$= \left(\frac{\partial \dot{f}}{\partial q}\frac{\partial g}{\partial p} - \frac{\partial g}{\partial q}\frac{\partial \dot{f}}{\partial p}\right) + \left(-\frac{\partial \dot{g}}{\partial q}\frac{\partial f}{\partial p} + \frac{\partial \dot{g}}{\partial p}\frac{\partial f}{\partial q}\right)$$

$$= \cancel{dt}\,[\dot{f}, g] + \cancel{[\dot{g}, f]}\,[f, \dot{g}]$$

$$\boxed{\frac{d}{dt}[f, g] = \left[\frac{df}{dt}, g\right] + \left[\cancel{\frac{dg}{dt}}\, f, g\,\frac{dg}{dt}\right]} \quad \text{(Ans)}$$

② (i) $G = G(p, q)$.

$$\Rightarrow \frac{dG}{dt} = \frac{\partial G}{\partial q}\cdot\dot{q} + \frac{\partial G}{\partial p}\dot{p} \cancel{+ \frac{\partial G}{\partial t}} \quad \text{—①}$$

$$[G, H] = \frac{\partial G}{\partial q}\frac{\partial H}{\partial p} - \frac{\partial G}{\partial p}\frac{\partial H}{\partial q} \quad \text{—(ii)}$$

∴ from Hamilton's EOM: $\dot{p} = -\frac{\partial f}{\partial q}$; $\dot{q} = \frac{\partial f}{\partial p}$. —(iii)

$$[G, H] = \frac{\partial G}{\partial q}\dot{q} + \frac{\partial G}{\partial p}\dot{p} \quad \text{—(iv)}$$ [using (iii) & (ii) in]

Using (iv) in ①: $\frac{dG}{dt} = [G, H] \cancel{+ \frac{\partial G}{\partial t}}$

$\frac{dG}{dt} = 0$ [$G = G(q, p)$ is a constant of motion]

$\therefore \boxed{[G, H] = 0}$ (Proved)

(B) $[F_1, H] = \sum_{i=1}^{2} \left(\frac{\partial F_1}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial F_1}{\partial p_i} \frac{\partial H}{\partial q_i} \right)$

$\therefore \frac{\partial F_1}{\partial p_1} = \frac{1}{a_2}$; $\frac{\partial F_1}{\partial q_1} = -\frac{a}{a_2}$

$\therefore \frac{\partial F_1}{\partial p_2} = 0$; $\frac{\partial F_1}{\partial q_2} = \frac{-(p_1 - aq_1)}{a_2^2}$

$\therefore \frac{\partial H}{\partial q_1} = p_1 - 2aq_1$; $\frac{\partial H}{\partial q_2} = -p_2 + 2bq_2$

$\therefore \frac{\partial H}{\partial p_1} = q_1$; $\frac{\partial H}{\partial p_2} = -q_2$

$\therefore \frac{\partial F_1}{\partial q_1} \cdot \frac{\partial H}{\partial p_1} - \frac{\partial F_1}{\partial p_1} \frac{\partial H}{\partial q_1} + \frac{\partial F_1}{\partial q_2} \frac{\partial H}{\partial p_2} - \frac{\partial F_1}{\partial p_2} \frac{\partial H}{\partial q_2}$

$= -\frac{a}{a_2} \cdot q_1 - \frac{1}{a_2}(p_1 - 2aq_1) + \frac{(p_1 - aq_1)}{a_2^2}$

$= -\frac{aq_1}{a_2} - \cancel{\frac{p_1}{a_2}} + \frac{2aq_1}{a_2} + \cancel{\frac{p_1}{a_2}} - \frac{aq_1}{a_2} = 0$

$\therefore \boxed{[F_1, H] = 0}$ (Proved)

$\therefore [F_2, H] = \sum_{i=1}^{2} \left(\frac{\partial F_2}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial F_2}{\partial p_i} \frac{\partial H}{\partial q_i} \right)$

$\therefore \frac{\partial F_2}{\partial q_1} = q_2$; $\frac{\partial H}{\partial p_1} = q_1$

$\frac{\partial F_2}{\partial p_1} = 0$; $\frac{\partial H}{\partial q_1} = p_1 - 2aq_1$

$\therefore \frac{\partial F_2}{\partial q_2} = q_1$; $\frac{\partial H}{\partial p_2} = -q_2$

$\frac{\partial F_2}{\partial p_2} = 0$; $\frac{\partial H}{\partial q_2} = -p_2 + 2bq_2$

$\therefore [F_2, H] = q_1 q_2 - q_1 q_2 = 0$ (Proved) $\boxed{[F_2, H] = 0}$

$\therefore \frac{dF_1}{dt} = \frac{dF_2}{dt} = 0$ [from Hamiltonian's EOM]

$\longrightarrow$ F_1 & F_2 are conserved (Proved)

Let $x = H$; $y = F_1$; $z = F_2$.

$\therefore [H, [F_1, F_2]] + [F_1, \overset{\to 0}{[F_2, H]}] + [F_2, \overset{\to 0}{[H, F_1]}] = 0$

$[\because [F_1, H] = 0$
$[F_2, H] = 0]$

$\Rightarrow [H, [F_1, F_2]] = 0.$

$[F_1, F_2]$ is a conserved qty.

$$\therefore [F_1, F_2] = \sum_{i=1}^{2} \left(\frac{\partial F_1}{\partial q_i} \frac{\partial F_2}{\partial p_i} - \frac{\partial F_1}{\partial p_i} \frac{\partial F_2}{\partial q_i} \right)$$

$$= -\frac{a}{q_2} \cdot 0 - \frac{1}{q_2} \cdot q_2 - \left(p_1 - \frac{a q_1}{q_2^2} \right) \cdot 0 - 0 \cdot q_1$$

$$= -1 \, a.$$

$[F_1, F_2] = -1 \rightarrow$ trivial. Hence no other conserved quantities. (Ans)

(3) (i) $T = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2)$. (in spherical polar coordinates)

$$\boxed{L = T - V = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) + \frac{k}{r}}$$

For r

$\therefore \frac{\partial L}{\partial r} = m r \dot{\theta}^2 - \frac{k}{r^2}$; $\frac{\partial L}{\partial \dot{r}} = m \dot{r}$

$\therefore \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{r}} \right) - \frac{\partial L}{\partial r} = 0$ [LEOM].

$\Rightarrow m \ddot{r} = m r \dot{\theta}^2 - \frac{k}{r^2} = -\frac{\partial V_{eff}}{\partial r}$ — (I)

$\therefore \frac{\partial V_{eff}}{\partial r} = -m r \dot{\theta}^2 + \frac{k}{r^2}$ — (I)

$\Rightarrow$ ~~V_{eff}~~ =

For θ:

$\frac{\partial L}{\partial \theta} = 0$; $\frac{\partial L}{\partial \dot{\theta}} = m r^2 \dot{\theta}$

$\therefore \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = 0 \Rightarrow \frac{\partial L}{\partial \dot{\theta}} = m r^2 \dot{\theta} = \text{constant} = L_z$

$\Rightarrow \dot{\theta} = \frac{L_z}{m r^2} = \frac{h}{r^2}$

$\Rightarrow \boxed{\dot{\theta}^2 = \frac{h^2}{r^4}}$ — (II)

Substituting (II) in (I):

$$\frac{\partial V_{eff}}{\partial r} = -mr \cdot \frac{h^2}{r^4} + \frac{k}{r^2}$$

$$= -\frac{mh^2}{r^3} + \frac{k}{r^2}.$$

$$\Rightarrow \boxed{V_{eff} = \frac{mh^2}{2r^2} - \frac{k}{r}} \quad \text{— (III)}$$

$$\therefore \boxed{\begin{aligned} m\ddot{r} &= \frac{mh^2}{r^3} - \frac{k}{r^2} \\ &= -\frac{\partial V_{eff}}{\partial r} \end{aligned}} \quad \text{— (IV)}$$

— Radial Eqn of Motion. (Ans)

(II) $L = L_0 + L_1 + L_2$.

Here, $L_1 = 0$; $L_0 = -V$; $L_2 = T$.

$\therefore h = T + V = E$.

$\therefore E = \frac{1}{2} m(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{k}{r}$.

$$\Rightarrow \frac{2}{m}\left(E + \frac{k}{r}\right) = \dot{r}^2 + r^2\dot{\theta}^2$$

$$\Rightarrow \dot{r}^2 = \frac{2}{m}\left(E + \frac{k}{r}\right) - r^2 \cdot \frac{h^2}{r^4} \quad \text{(from (II))}$$

$$\boxed{\dot{r}^2 = \frac{2}{m}\left(E + \frac{k}{r}\right) - \frac{h^2}{r^2}} \quad \text{— (V)}$$

@ r_{min} & r_{max}:

$$\dot{r} = 0 \Rightarrow \frac{2}{m}\left(E + \frac{k}{r}\right) = \frac{h^2}{r^2}$$

$$\Rightarrow Er^2 + kr = \frac{mh^2}{2}$$

$$\Rightarrow r^2 + \frac{k}{E}r - \frac{mh^2}{2E} = 0$$

$$\therefore r = \frac{-\frac{k}{E} \pm \sqrt{\frac{k^2}{E^2} + \frac{2mh^2}{E}}}{2} = \frac{k}{2E} \pm \frac{1}{2}\sqrt{\frac{k^2}{E^2} + \frac{2L_2^2}{m^2E}}$$

$$\therefore r_{max} = \frac{k}{2E} + \frac{1}{2}\sqrt{\frac{k^2}{E^2} + \frac{2L_2^2}{mE}}$$

$$r_{min} = \frac{k}{2E} - \frac{1}{2}\sqrt{\frac{k^2}{E^2} + \frac{2L_2^2}{mE}}$$

[Ans]

(iii) $a = \frac{r_{min} + r_{max}}{2} = \frac{k}{2E}$ (5)

a depends only on the energy of the particle

(iv) (a) E can be used to control the size of the orbit — ~~se~~ a

(b) ~~Ro~~ From solving the orbit eqn, we get:

$e^2 = f(L_z, E)$; e – eccentricity of orbit.

Thus, L_z & E can be used to change the eccentricity (shape) of the orbit.

(5) Let the canonical variables be (x, y, z) & (p_x, p_y, p_z).

$\therefore \bar{L} = \bar{r} \times \bar{p}$

$\therefore L_x = y p_z - z p_y$; $L_y = z p_x - x p_z$; $L_z = x p_y - y p_x$

(a) $\{L_x, L_y\} = \sum_i \left(\frac{\partial L_x}{\partial q_i} \frac{\partial L_y}{\partial p_i} - \frac{\partial L_x}{\partial p_i} \frac{\partial L_y}{\partial q_i} \right)$

Now, $\frac{\partial L_x}{\partial x} = 0$; $\frac{\partial L_x}{\partial y} = p_z$; $\frac{\partial L_x}{\partial z} = -p_y$

$\frac{\partial L_y}{\partial x} = -p_z$; $\frac{\partial L_y}{\partial y} = 0$; $\frac{\partial L_y}{\partial z} = p_x$

$\therefore$ ~~$\{L_x, L_y\}$~~ $\frac{\partial L_x}{\partial p_x} = 0$; $\frac{\partial L_x}{\partial p_y} = -z$; $\frac{\partial L_x}{\partial p_z} = y$

$\frac{\partial L_y}{\partial p_x} = z$; $\frac{\partial L_y}{\partial p_y} = 0$; $\frac{\partial L_y}{\partial p_z} = -x$

$\therefore \{L_x, L_y\} = (-p_y)(-x) - y p_x$

$= x p_y - y p_x = L_z$ (Ans)

By similar calculations:

$\{L_y, L_z\} = L_x$; $\{L_z, L_x\} = L_y$ (Ans)

$\therefore \boxed{\{L_i, L_j\} = \epsilon_{ijk} L_k}$ (Ans) [$i = 1, 2, 3$ ~~x, y, z~~]